

Abstract

This paper proposes a novel method for retrieval augmented generation. We show that our method improves faithfulness by 15 percent over baselines.

1. Introduction

Retrieval augmented generation combines retrieval with generation. Hallucination remains a key challenge in these systems. In this work we address this problem directly using a verification loop.

2. Method

Our method uses a two-stage pipeline: generation followed by verification. The verifier checks each claim against retrieved source chunks. If a claim cannot be verified, it is flagged as unsupported.

3. Experiments

We evaluate on a dataset of 50 research papers with hand labeled Q&A pairs. Our faithfulness score improved from 0.72 to 0.91 after adding verification.

References

- [1] Smith et al. 2023. Retrieval methods.
- [2] Doe et al. 2024. Verification loops in RAG.